

DFI 5.2 Back-Propagation Analysis

Memory Controller Signal Reference & Internal Functional Blocks

Based on DDR PHY Interface Specification v5.2 (October 2024) • Covers all 11 DFI interface groups • Full MC back-propagation to AXI ports

Back-Propagation Method: Starting from each DFI signal on the MC side of the interface, this document traces what internal MC functional block drives or consumes that signal, what DRAM operation it maps to, and ultimately how it connects back to the AXI subordinate port. The final page shows a complete block diagram of all functional blocks from AXI inward to the DFI boundary.

Interface Group 1 — Command Interface

MC → PHY (except dfi_alert_n which is PHY → MC)

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_address_pN	M→P	14b	B1	Row/column address or CA bus value. Driven by the Command/Address Encoder after the scheduler selects an ACTIVATE or CAS command. Maps to DRAM A[13:0] pins via the PHY.
dfi_act_n_pN	M→P	1b	B1	DDR4/5 explicit ACTIVATE encoding. Low = ACTIVATE command. Replaces the legacy RAS/CAS/WE decode used in DDR3 and earlier. Driven simultaneously with bank and address signals.
dfi_2n_mode_pN	M→P	1b	B1	DDR5 2N command mode enable. When asserted, the command window is doubled to improve signal integrity margins at high transfer rates. Programmed during init FSM.
dfi_alert_n_aN	P→M	1b	B13	Reverse direction. PHY reports a CA parity error or DDR4/5 CRC write error back to the MC. Consumed by the CA Parity & CRC Engine which decides whether to retry the command.
dfi_bank_pN	M→P	3b	B1	Bank address (BA[2:0]) targeting which of the 8 DRAM banks is addressed. Always presented together with the row/column address by the Timing FSM.
dfi_bg_pN	M→P	2b	B1	Bank group address (DDR4/5). Selects which of 4 bank groups the transaction targets, enabling bank group parallelism and back-to-back CAS commands with tCCDS instead of tCCDL.
dfi_cas_n_pN	M→P	1b	B1	CAS command (DDR1–4 legacy). Combined with RAS_n and WE_n to encode the JEDEC command truth table (ACTIVATE, READ, WRITE, PRECHARGE, REFRESH).
dfi_cid_pN	M→P	3b	B1	Chip ID for 3D-stacked DDR4/5 LRDIMM/3DS configurations. Selects which logical die within a stacked package is addressed. Driven by scheduler when multiple die-select is active.
dfi_cke_pN	M→P	1b	B8	Clock enable. Driven by the Init FSM during power-up sequencing and by the Power Management Unit to gate the DRAM clock for self-refresh or power-down entry/exit.
dfi_cs_pN	M→P	1b	B1	Chip select. Asserted by the Timing FSM to activate the targeted rank for each command. In DDR5, CS is also used for command encoding (2-cycle command).
dfi_cs_geardown_pN	M→P	1b	B1	DDR5 geardown mode chip select. Separate CS path used when geardown mode is enabled, reducing command bus switching activity for signal integrity at high speeds.

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_dram_clk_disable_pN	M→P	1b	B11	Instructs PHY to stop driving the differential CK/CKn signals to DRAM. Driven by Power Management Unit during self-refresh and deep power-down to eliminate clock switching power.
dfi_odt_pN	M→P	1b	B16	On-Die Termination control. The ODT Timing Controller computes the exact assertion and de-assertion cycles (tAOND/tAOFD) relative to READ and WRITE commands across all ranks.
dfi_parity_in_pN	M→P	1b	B13	CA parity bit computed by the CA Parity & CRC Engine over the address/command bus and forwarded through the PHY to the DRAM for error detection. Used in DDR4/5 and RDIMM systems.
dfi_ras_n_pN	M→P	1b	B1	RAS command (DDR1–4 legacy). Part of the 3-bit command encoding: RAS_n, CAS_n, WE_n. Not present in DDR5 which uses the CA bus and ACT_n instead.
dfi_reset_n_pN	M→P	1b	B8	DRAM reset signal. Held low during power-on reset sequence. Released by the Init FSM according to the JEDEC initialisation sequence timing (tINIT5 for DDR4, etc.).
dfi_we_n_pN	M→P	1b	B1	Write enable (DDR1–4 legacy). Third component of the command truth table alongside RAS_n and CAS_n.

Interface Group 2 — Write Data Interface

MC → PHY

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_wrdata_pN	M→P	DQ×2	B5	Actual write data payload. DDR double-edge packed so width = 2× physical DQ bus per phase. Sourced from the Write Data Buffer which received beats from the AXI W channel. Presented tphy_wrlat DFI cycles after the WRITE command.
dfi_wrdata_crc_pN	M→P	varies	B13	Write CRC check symbols appended to the write data burst. Computed by the CRC Engine when DDR4/5 write CRC mode is enabled. Transmitted alongside wrdata in the same DFI cycle.
dfi_wrdata_cs_pN	M→P	1b	B5	Write data rank qualifier. Associates data beats with a specific rank in multi-rank systems, allowing the PHY to correctly gate the DQS strobe per rank during write bursts.
dfi_wrdata_ecc_pN	M→P	varies	B5	Link ECC sideband symbols for LPDDR5 link ECC mode. When phy_linkecc_mode=1, ECC symbols ride alongside data on a separate bus rather than replacing data bits.
dfi_wrdata_en_pN	M→P	1b	B5	Write data enable strobe. Tells PHY exactly which DFI cycles to drive DQ outputs and assert DQS. PHY uses this to enable output drivers and control DQS gating precisely. Timing is defined by tphy_wrlat and tphy_wrdata parameters.
dfi_wrdata_mask_pN	M→P	DQ/8	B5	Byte-lane write mask (DDR DM signal). One bit per byte lane. MC asserts individual bits to suppress specific byte lanes from being written to DRAM. Derived from AXI WSTRB after width adaptation. In DDR4/LPDDR4 write DBI mode this signal becomes the write DBI signal instead.

Interface Group 3 — Read Data Interface

MC ↔ PHY (bidirectional)

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_rddata_en_pN	M→P	1b	B6	Read data enable. MC asserts this trddata_en DFI cycles after issuing a READ command. This arms the PHY's DQS gate capture logic to sample incoming DQ data. Without this signal PHY input buffers stay disabled, saving power.
dfi_rddata_wN	P→M	DQ×2	B6	Captured read data returned from PHY to MC. Width is 2× DQ bus per frequency-ratio word. Consumed by the Read Data Return Path which un-packs beats back to AXI data width and routes them via the R channel.
dfi_rddata_crc_wN	P→M	varies	B13	CRC check bits extracted from the DRAM read bus, returned alongside rddata. Consumed by the CRC Engine in the MC for read CRC verification. Any mismatch triggers the Error Handler.
dfi_rddata_cs_pN	M→P	1b	B6	Read data chip select qualifier. Tells PHY which rank the in-flight read belongs to. PHY uses this to correctly apply per-rank DQS gating delays, since different ranks have different read latencies.
dfi_rddata_dbi_wN	P→M	DQ/8	B6	Data Bus Inversion status bits returned by PHY. MC must un-invert the corresponding byte lanes where DBI was applied by the DRAM. One bit per byte lane. Only applicable for DDR4, LPDDR4, and LPDDR5.
dfi_rddata_dnv_wN	P→M	1b	B6	Data Not Valid flag (LPDDR2 specific). PHY signals that certain read data words within the burst are invalid due to the LPDDR2 variable latency gap. MC discards flagged words.
dfi_rddata_valid_wN	P→M	1b	B6	The fundamental read data valid strobe. PHY asserts this when captured read data on dfi_rddata is reliable, aligned, and ready to be latched. MC's Read Buffer uses rising edge to capture data into the AXI return pipeline.

Interface Group 4 — Update Interface

MC ↔ PHY (handshake)

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_ctrlupd_req	M→P	1b	B9	MC-initiated update request. MC asserts this to place the DFI bus in idle state before a self-refresh exit or any operation requiring PHY recalibration. Bus Arbiter must drain all in-flight transactions first. Required before every self-refresh exit per spec.
dfi_ctrlupd_ack	P→M	1b	B9	MC-initiated update acknowledge. PHY confirms DFI bus is idle and it is ready to proceed. Must assert before tctrlupd_min cycles. While asserted, bus must remain idle except for update-specific transactions.
dfi_phyupd_req	P→M	1b	B9	PHY-initiated update request. PHY requests DFI bus idle for its own internal operation (DLL re-lock, impedance calibration, read/write training maintenance). MC must respond within tphyupd_resp cycles.
dfi_phyupd_ack	M→P	1b	B9	PHY-initiated update acknowledge. MC confirms all in-flight AXI transactions have completed, no new commands will be issued. MC holds this asserted until PHY de-asserts dfi_phyupd_req.

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_phyupd_type	P→M	2b	B9	PHY update type selector. Encodes which of 4 PHY update operation types is being requested (0x0-0x3). Each type maps to a different maximum duration (tphyupd_type0 through tphyupd_type3). MC uses this to bound how long it must idle the bus.

Interface Group 5 — Status Interface

MC → PHY (except dfi_init_complete)

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_init_start	M→P	1b	B8	Init/frequency-change trigger. Asserted by Init FSM during power-up to signal that dfi_frequency, dfi_freq_fsp, dfi_cmd_freq_ratio and dfi_data_freq_ratio are valid. Also asserted at runtime to request a DRAM frequency change (LPDDR4/5 FSP switch).
dfi_init_complete	P→M	1b	B8	PHY ready signal. PHY asserts when initialisation or frequency change is complete. MC must not issue any DFI transactions until this is high. Init FSM waits on this before releasing normal operation to the scheduler.
dfi_frequency	M→P	6b	B8	Frequency index (0-63). MC tells PHY which of its pre-configured frequency profiles to activate. PHY uses this to reconfigure PLL, DLL, and I/O timing parameters for the new operating point.
dfi_freq_fsp	M→P	1-2b	B8	Frequency Set Point selector (LPDDR4/5). Indicates which FSP register set in the DRAM MR is currently active. PHY uses this to apply the correct per-FSP timing parameters stored during training.
dfi_cmd_freq_ratio	M→P	2b	B8	Command clock ratio: 00=1:1, 01=1:2, 10=1:4. Tells PHY how many command phases are packed per DFI clock cycle. Determines the _pN suffix multiplicity on command signals.
dfi_data_freq_ratio	M→P	2b	B8	Data clock ratio. Same encoding as cmd_freq_ratio but applies to the data path. Controls how many data words (_wN) are packed per DFI cycle.

Interface Group 6 — PHY Managed Interface

PHY → MC request / MC → PHY acknowledge

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_phymngd_req	P→M	1b	B10	PHY requests full control of the DRAM bus. PHY will run an autonomous sequence (ZQ calibration, on-die training, PPR) without MC involvement. MC must respond within tphymngd_resp cycles.
dfi_phymngd_ack	M→P	1b	B10	MC grants bus control to PHY. MC has drained all in-flight transactions and put the DRAM in the state specified by dfi_phymngd_cs_state. Held asserted until PHY de-asserts dfi_phymngd_req.
dfi_phymngd_type	P→M	varies	B10	Operation type encoding. Tells the MC what the PHY intends to do. The MC uses this to know how long to remain idle and whether it needs to issue any commands (e.g., refresh) before handing over.
dfi_phymngd_cs_state	P→M	1b/rank	B10	Required DRAM state per chip select. PHY specifies what state it requires each rank to be in before handover (active, power-down, self-refresh, etc.). MC is responsible for achieving that state if dfi_phymngd_state_sel is set accordingly.
dfi_phymngd_state_sel	P→M	1b	B10	State responsibility selector. Determines whether the MC or the PHY is responsible for placing DRAM in the required state before the PHY takes control. When MC is responsible, it issues the necessary commands before asserting ack.

Interface Group 7 — Low Power Control Interface

MC → PHY request / PHY → MC acknowledge

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_lp_ctrl_req	M→P	1b	B11	MC signals no more commands will arrive on the command interface. PHY may power down its command-path receivers, PLLs, and clock distribution to the command bus. Asserted by Power Management Unit after transaction activity goes idle.
dfi_lp_ctrl_ack	P→M	1b	B11	PHY acknowledges low-power entry on the control path. Must assert within tlp_resp cycles of dfi_lp_ctrl_req. After assertion the PHY command path may be asleep.
dfi_lp_ctrl_wakeup	M→P	6b	B11	Control path wakeup latency budget. MC tells PHY the maximum allowed wakeup time. PHY uses this to select the deepest sleep state it can exit within that budget. 16 predefined wakeup times from 16 to 262144 DFI clock cycles.
dfi_lp_data_req	M→P	1b	B11	Same as ctrl_req but for the data interface. No more read or write data transfers are pending. PHY may power down DQ/DQS output drivers, input receivers, and data-path DLLs.
dfi_lp_data_ack	P→M	1b	B11	PHY acknowledges data-path low-power entry. After this the PHY data path may be asleep until woken by de-assertion of dfi_lp_data_req.
dfi_lp_data_wakeup	M→P	6b	B11	Data path wakeup latency budget. Same encoding as ctrl_wakeup but applies to the data-path circuits.

Interface Group 8 — Error Interface

PHY → MC

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_error	P→M	1b	B12	PHY-detected error indicator. Asserted within terror_resp cycles of the DFI transaction known to be affected. Error types include DRAM ECC failure, DLL loss of lock, read DQS training error, link CRC error. Consumed by RAS/Error Handler.
dfi_error_info	P→M	4b	B12	Error type encoding. Identifies which error category occurred so the RAS/Error Handler can log it, propagate status to the AXI RUSER/BUSER sideband, trigger a scrub, or initiate a PHY managed re-training sequence.

Interface Group 9 — MC to PHY Message Interface

MC → PHY out-of-band messaging

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_ctrlmsg_req	M→P	1b	B14	Message request. MC asserts to begin an out-of-band message transaction. Signals to the PHY that the ctrlmsg and ctrlmsg_data signals are valid.
dfi_ctrlmsg	M→P	8b	B14	Message opcode. Encodes which PHY action is being requested by the MC (e.g., trigger ZQ calibration, modify ODT configuration, request specific training step, change power mode). Valid only while dfi_ctrlmsg_req is asserted.
dfi_ctrlmsg_data	M→P	16b	B14	Message payload data. Parameter information accompanying the opcode. PHY interprets this in the context of the ctrlmsg opcode. Valid only while dfi_ctrlmsg_req is asserted.
dfi_ctrlmsg_ack	P→M	1b	B14	Message acknowledge. PHY confirms it received and processed the message. Must assert within tctrlmsg_resp cycles. De-asserts within tctrlmsg_max cycles after assertion.

Interface Group 10 — Disconnect Protocol

MC → PHY

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_disconnect_error	M→P	1b	B9	DFI disconnect error flag. MC asserts during an update or PHY managed handshake to signal an error condition requires disconnecting the PHY. Tells PHY not to wait for the normal handshake completion sequence. PHY must disconnect within tc_trlupd_disconnect_error or tphyrngd_disconnect_error cycles.

Interface Group 11 — WCK Control Interface

MC → PHY (LPDDR5 / LPDDR5X only)

Signal	Dir	Width	MC Block	Description & Back-Propagation
dfi_wck_cs_pN	M→P	1b	B15	WCK chip select qualifier. Associates the WCK control signals with a specific rank/channel in multi-rank LPDDR5 systems. Driven by the WCK Controller in sync with write data.
dfi_wck_en_pN	M→P	1b	B15	WCK output enable. MC gates the WCK clock output from PHY to DRAM when no write data is imminent. Disabling WCK when idle saves significant dynamic power in LPDDR5 systems.
dfi_wck_toggle_pN	M→P	1b	B15	WCK toggle mode selector. Selects between fast-toggle (continuous 2× rate during writes) and slow/static mode during idle and read periods. LPDDR5 JEDEC requires fast toggle during write operations. Driven by WCK Controller coordinated with the write data enable path.

MC Block Reference: B1=Cmd/Addr Encoder & Timing FSM B2=Open-Row Tracker B3=Command Scheduler B4=Transaction Queue & Address Decoder B5=Write Data Buffer & Packer B6=Read Data Return Path B7=Refresh Engine B8=Init FSM & Freq Change Controller B9=DFI Bus Arbiter / Update Controller B10=PHY Managed Interface Controller B11=Power Management Unit B12=RAS / Error Handler B13=CA Parity & CRC Engine B14=MC-to-PHY Message Engine B15=WCK Controller (LPDDR5) B16=ODT Timing Controller B17=AXI Subordinate Port(s)

Memory Controller — Internal Functional Blocks

Back-propagated from DFI 5.2 MC ports through to AXI subordinate interface

B1 — Command/Address Encoder & Timing FSM

Receives commands from B3 (scheduler). Encodes them as DFI command signals: `dfi_address`, `dfi_act_n`, `dfi_bank`, `dfi_bg`, `dfi_cas_n`, `dfi_ras_n`, `dfi_we_n`, `dfi_cs`, `dfi_cke`, `dfi_odt`, `dfi_reset_n`, `dfi_parity_in`, `dfi_cid`, `dfi_2n_mode`, `dfi_cs_geardown`, `dfi_dram_clk_disable`. Enforces all JEDEC timing constraints (`tRCD`, `tRP`, `tRAS`, `tRRD`, `tFAW`, `tWTR`, `tRC`, `tCCD`, `tCCDL`, `tCCDS`) in hardware before asserting each signal. Generates command-phase replication for frequency-ratio systems (`_pN` suffix).

B2 — Row/Bank/Rank State Machine (Open-Row Tracker)

Maintains a state table of which row is currently open in each bank across all ranks (typically 8–16 entries per bank group per rank). On each incoming request determines: row-buffer HIT (CAS directly), row-buffer MISS (PRECHARGE + ACTIVATE + CAS), or row-buffer EMPTY (ACTIVATE + CAS). Feeds hit/miss status to B3. Prevents illegal bank state transitions. Updated on every ACT, PRE, and REF command.

B3 — Command Scheduler / Reorder Engine

Receives read/write requests from B4. Reorders them for maximum row-buffer locality while respecting QoS priorities from the AXI AWQOS/ARQOS signals. Arbitrates across multiple AXI ports. Implements scheduling policy (FR-FCFS, FR-FCFS with priority banding, or deadline-aware). Feeds decisions to B1 (command encoding) and B5/B6 (data path). Pauses during update (B9), refresh (B7), init (B8), and PHY managed (B10) windows.

B4 — Transaction Queue & Address Decoder

Accepts AXI transactions from AW and AR channels of B17. Decodes AXI byte address into DRAM coordinates: {rank, bank group, bank, row, column} using the configured DRAM address mapping (interleaved, sequential, or XOR-mapped). Buffers pending transactions with their AXI IDs and QoS tags. Manages AXI ID ordering rules. Passes decoded requests to B3. Feeds write data from B5 and read responses back to B17.

B5 — Write Data Buffer & Packer

Receives AXI W-channel beats from B17. Packs them to DFI data width (e.g., 512 b for DDR5 BL16, 256 b for DDR4 BL8). Converts AXI WSTRB to `dfi_wrddata_mask` byte-lane enables. Schedules presentation of `dfi_wrddata` and `dfi_wrddata_en` at the correct DFI cycle offset (`tphy_wrlat` + `tphy_wrddata`) after the WRITE command. Appends CRC symbols if enabled (B13). Generates `dfi_wrddata_cs` for rank qualification and `dfi_wrddata_ecc` for LPDDR5 link ECC.

B6 — Read Data Return Path & Unpacker

Monitors `dfi_rddata_valid` from PHY. On valid assertion, latches `dfi_rddata` into a return buffer. Un-inverts DBI bytes using `dfi_rddata_dbi`. Discards words flagged by `dfi_rddata_dnv` (LPDDR2). Unpacks DFI-width data back to AXI beat width. Routes read data to B17 R channel with the correct AXI RID matching the original ARID. Triggers ECC correction logic in B12 if ECC is enabled. Generates `dfi_rddata_en` and `dfi_rddata_cs` at the correct pre-computed cycle offset.

B7 — Refresh Engine

Maintains a `tREFI` countdown for each rank. On expiry, signals B3 to insert an AUTO REFRESH command. Handles deferred refresh: DDR5 allows up to 8 consecutive refresh postponements (`tREFI8`) before a mandatory refresh window must be served. Manages per-bank refresh (PBR) in DDR5 if enabled. Drives the necessary `dfi_cs` assertion and refresh command encoding via B1. Also manages self-refresh entry/exit sequencing in coordination with B8 and B11.

B8 — Init FSM & Frequency Change Controller

Sequences the complete DRAM power-up protocol: assert `dfi_reset_n` low, wait `tINIT`, ramp `dfi_cke`, ZQCL calibration, Mode Register writes (MR0–MR6 for DDR4, MR0–MR13 for DDR5). Drives `dfi_init_start`, waits for `dfi_init_complete`. Drives `dfi_frequency`, `dfi_freq_fsp`, `dfi_cmd_freq_ratio`, `dfi_data_freq_ratio`. Also handles runtime frequency switching for LPDDR4/5 FSP changes and DDR5 frequency change protocol.

B9 — DFI Bus Arbiter / Update Controller

Manages the Update Interface handshake: `dfi_ctrlupd_req/ack` (MC-initiated) and `dfi_phyupd_req/ack` (PHY-initiated). Drains all in-flight DFI transactions before asserting any update request. Gates B3 during update windows. Monitors `tctrlupd_interval` to meet mandatory update periodicity. Also drives `dfi_disconnect_error` (B10/Disconnect Protocol) when an error forces abnormal handshake termination. Implements the full disconnect protocol per DFI 5.2 Section 4.21.

B10 — PHY Managed Interface Controller

Handles the `dfi_phymngd_req/ack` handshake. On receiving `dfi_phymngd_req`, decodes `dfi_phymngd_type` and `dfi_phymngd_cs_state`. If MC is responsible for DRAM state (per `dfi_phymngd_state_sel`), issues the required commands via B1 before asserting `ack`. Holds B3 paused for the entire PHY managed window. Resumes normal operation after PHY releases the bus.

B11 — Power Management Unit

Monitors transaction activity across all AXI ports. When the command and data paths are idle beyond a programmable threshold, asserts `dfi_lp_ctrl_req` and `dfi_lp_data_req` with appropriate `dfi_lp_ctrl_wakeup/dfi_lp_data_wakeup` values. Manages coordinated self-refresh entry and exit. Also drives `dfi_dram_clk_disable` during deep power modes. Wakes up the system when a new AXI transaction arrives, issuing the correct wakeup latency sequence before releasing B3.

B12 — RAS / Error Handler

Receives `dfi_error` and `dfi_error_info` from PHY. Decodes error type (ECC, DLL, DQS, CRC, link error). Logs the error with associated AXI transaction ID and timestamp. Propagates correctable ECC status to the AXI R channel RUSER field and uncorrectable status to BRESP/RRESP. May trigger a scrub sequence via B3 or initiate a PHY re-training request via B14 or B10.

B13 — CA Parity & CRC Engine

Computes even/odd parity over the command/address bus, drives `dfi_parity_in`. Monitors `dfi_alert_n` for error reports from DRAM/PHY. For write CRC: computes CRC-8 over write data bursts, appends to `dfi_wrdata_crc`. For read CRC: verifies `dfi_rddata_crc` received from PHY. Any CRC mismatch is reported to B12. Manages the CRC retry protocol if the DRAM asserts `alert_n`.

B14 — MC-to-PHY Message Engine

Drives `dfi_ctrlmsg_req`, `dfi_ctrlmsg` (opcode), and `dfi_ctrlmsg_data` (payload). Used to send out-of-band control messages to PHY firmware without occupying the main command bus. Typical messages: trigger ZQ calibration, modify ODT settings, request specific training sequence, change clock gating policy. Waits for `dfi_ctrlmsg_ack` before releasing the message channel.

B15 — WCK Controller (LPDDR5 / LPDDR5X only)

Manages `dfi_wck_en`, `dfi_wck_cs`, and `dfi_wck_toggle`. Coordinates write-clock gating with the write data path (B5). Enables fast-toggle mode before write bursts and switches to slow/static during reads and idle periods to save power. WCK runs at 2× the DRAM data rate; it is entirely PHY-internal signal infrastructure, but the MC controls its gating and mode via these DFI signals.

B16 — ODT Timing Controller

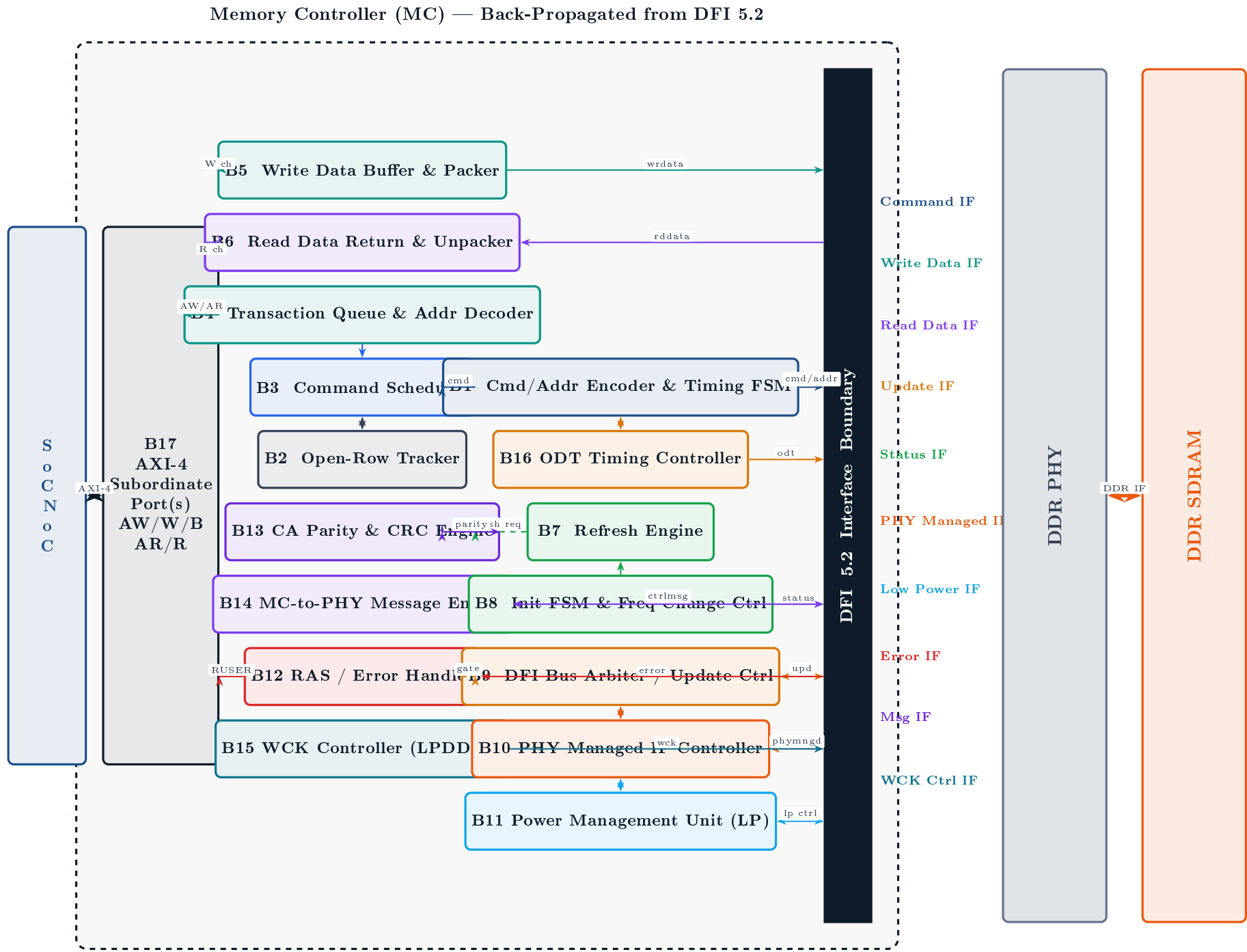
Computes the exact cycle-precise assertion and de-assertion of `dfi_odt_pN` for each rank. Must satisfy JEDEC tAOND (ODT turn-on delay), tAOFD (ODT turn-off delay), tAON (ODT turn-on), and tAOF (ODT turn-off) relative to READ and WRITE commands across all active and non-target ranks simultaneously. Particularly complex in multi-rank systems where ODT must be applied to non-targeted ranks during writes to dampen signal reflections.

B17 — AXI4 Subordinate Port(s)

The outermost MC interface. Accepts AXI4 transactions from the SoC NoC/interconnect across 5 channels: **AW** (write address: AWVALID, AWREADY, AWID, AWADDR, AWLEN, AWSIZE, AWBURST, AWLOCK, AWCACHE, AWPROT, AWQOS, AWREGION, AWUSER), **W** (write data: WVALID, WREADY, WDATA, WSTRB, WLAST, WUSER), **B** (write response: BVALID, BREADY, BID, BRESP, BUSER), **AR** (read address: ARVALID, ARREADY, ARID, ARADDR, ARLEN, ARSIZE, ARBURST, ARLOCK, ARCACHE, ARPROT, ARQOS, ARREGION, ARUSER), **R** (read data: RVALID, RREADY, RID, RDATA, RRESP, RLAST, RUSER). Tracks outstanding IDs. Passes QoS to B3. Routes write data to B5, read data from B6, write responses from B5. Carries ECC and RAS status in BUSER and RUSER sideband fields.

Memory Controller — Complete Internal Block Diagram

Back-propagated from DFI 5.2 MC ports • Signal flow: AXI Subordinate $\Rightarrow$ Internal Blocks $\Rightarrow$ DFI 5.2 Interface



Block Reference: B1=Cmd/Addr Encoder & Timing FSM B2=Open-Row Tracker B3=Command Scheduler B4=Transaction Queue & Addr Decoder B5=Write Data Buffer B6=Read Data Return Path B7=Refresh Engine B8=Init FSM & Freq Change Ctrl B9=DFI Bus Arbiter / Update Ctrl B10=PHY Managed IF Ctrl B11=Power Management Unit B12=RAS / Error Handler B13=CA Parity & CRC Engine B14=MC-to-PHY Message Engine B15=WCK Controller (LPDDR5) B16=ODT Timing Ctrl B17=AXI-4 Subordinate Port(s)